

KSP INTELLIGENCE COMMAND

From Crime Records to Preventive Action

Hackathon Product, UX, Backend, AI & Analytics Blueprint

Consolidated from the discussion on case lifecycle, case health, focus subjects, intelligence, prediction, APIs, real-time operations and Catalyst architecture.

Core product idea

Do not build a single crime-statistics dashboard. Build a Karnataka Police Intelligence & Operations Platform that converts case data into actionable intelligence: **Observe** → **Understand** → **Prioritize** → **Act** → **Measure** → **Learn** → **Predict**.

The existing ERD is the system-of-record foundation. The proposed intelligence layer should sit above it, producing derived health signals, observations, predictions and intervention outcomes without unnecessarily altering the core police schema.

1. Product Vision

- State-wide decision support for senior command, district operations, investigation teams and field officers.
- The home page answers: What changed? What is dangerous? What is stuck? Where should we look? What should we do?
- The same underlying data should produce different workspaces for different roles.

2. Navigation / Workspaces

Workspace	Primary question
Command Center	What needs my attention now?
Crime Intelligence	What is happening and what changed?
Case Health	Are active investigations progressing appropriately?
Investigation	What has happened in this case and what is pending?
Hotspot Intelligence	Where should preventive resources focus?
People & Networks	Which cases/entities are connected?
Predictive Risk	Where/when might crime risk increase?
Live Operations	What is happening now?
Performance	Where are workload/process bottlenecks?
KSP Copilot	Ask, explain, compare, drill down and navigate using evidence.

3. Command Center — Awe-Factor Home

Karnataka State should be the starting context, with filters for period, district, unit, crime head/sub-head, case status and other supported dimensions.

- **KPI strip:** FIRs, active cases, cases under investigation, chargesheeted cases, attention signals, hotspots, silent cases and critical signals.

- **Attention Queue:** emerging crime clusters, investigation-health signals, location anomalies, repeat associations and other explainable observations.
- **Karnataka map:** crime density, change, hotspots, investigation concentration, clusters, police units and predictive risk as switchable layers.
- **Crime movement:** month-over-month and year-over-year changes rather than counts alone.
- **Case health:** healthy / review / stalled / critical as analytical categories for the prototype, not official KSP classifications unless validated.

4. Case Health — Key Differentiator

Case status is not the same as case health. Two cases can both be 'Under Investigation' while one is progressing and the other has become silent. Health should be an explainable combination of observable progress, age, activity, ownership/workload and unresolved milestones.

Dimension	What to observe
Progress	FIR, IO assignment, victim/complainant, accused, arrest, court/chargesheet and other available milestones.
Age	Time since registration, interpreted against configured review thresholds rather than invented legal deadlines.
Activity	Most recent recorded investigation activity; this requires an explicit activity/event model for robust measurement.
Ownership	Investigating officer, unit and district workload.
Investigation outcome	Chargesheet/final report concepts and current status.
Signals	Ageing, inactivity, missing expected information, unusual delay or other review-worthy patterns.

5. Recommended Intelligence-Layer Tables

Keep the existing police ERD as the system of record. Add a separate analytical/intelligence layer.

- **CaseActivity:** ActivityID, CaseMasterID, ActivityTypeID, ActivityDateTime, EmployeeID, Remarks.
- **ActivityTypeMaster:** investigation assigned, statement recorded, evidence collected, CCTV reviewed, suspect identified, arrest, recovery, forensic request/result, court submission, chargesheet preparation, supervisor review, etc.
- **CaseReview:** ReviewID, CaseMasterID, reviewer, review date, outcome and remarks.
- **AttentionSignal:** SignalID, CaseMasterID/subject, signal type, severity, score, reason, detected date, review status and reviewer.
- **CrimeCluster / Hotspot:** geographic cluster, crime type, period, trend, supporting cases and confidence.
- **Prediction:** target area/category, horizon, risk, confidence, model version and contributing factors.
- **Intervention / InterventionOutcome:** action taken, geography, period, intended target, before/after metrics and observed outcome.
- **AIQueryAudit:** user, question, tools/data consulted, answer metadata and audit information.

6. Focus Subjects

Subject	Questions
Case	What happened to this FIR/case? Is the investigation progressing?
Person / Entity	Which synthetic entities appear across cases? What links exist?
Location	Where is crime concentrating or changing?

Subject	Questions
Time	When are incidents concentrating?
Police Process	Where is work delayed, silent or overloaded?
Crime Pattern	What is increasing, decreasing or anomalous?
Intervention	Did an action appear to change the monitored outcome?

7. Prediction & Prevention

For the hackathon, focus predictive capability on places, crime categories, time windows and resource needs—not person-level claims about who will commit crime.

- 7-day / 14-day / 30-day risk views with confidence and contributing factors.
- Explainable drivers: recent increase, historical recurrence, temporal concentration, neighbouring hotspot, prior pattern.
- Use predictions as decision support; human officers remain responsible for decisions.
- Close the loop: Detect → Prioritize → Act → Measure → Learn → Predict.

8. Intervention Feedback Loop

A signature feature can connect a hotspot or pattern to an intervention and then measure the monitored outcome. Example: a theft hotspot is identified, a targeted patrol is simulated/deployed, and subsequent incidents are compared with a defined baseline. Avoid claiming causation unless the evaluation design supports it.

9. KSP Copilot

The AI assistant should be context-aware and evidence-grounded, not a generic chatbot. It should retrieve data through backend tools, run deterministic analytics, then use the LLM to explain the evidence.

```
USER → COPILOT → INTENT ROUTER → CASE / CRIME / GEO / HEALTH TOOLS → ZCQL / OLAP → ANALYTICS → EVIDENCE → LLM → EXPLAINABLE ANSWER
```

- Example: 'Why is Bengaluru red?' → retrieve trends, hotspots and signals → explain the top drivers.
- Example: 'Show cases behind this hotspot.' → open the filtered case view.
- Example: 'Which investigations need review?' → retrieve health signals and show evidence.
- Example: 'Summarize this case.' → produce a grounded timeline/status summary.

10. Backend API Catalogue

Dashboard

```
GET /api/dashboard/overview
GET /api/dashboard/kpis
GET /api/dashboard/attention
GET /api/dashboard/trends
GET /api/dashboard/district-summary
GET /api/dashboard/crime-summary
```

Cases

```
GET /api/cases
GET /api/cases/:caseId
GET /api/cases/:caseId/timeline
GET /api/cases/:caseId/health
GET /api/cases/:caseId/signals
GET /api/cases/:caseId/people
```

```
GET /api/cases/:caseId/legal
GET /api/cases/:caseId/location
GET /api/cases/:caseId/related
```

Case Health

```
GET /api/case-health/summary
GET /api/case-health/ageing
GET /api/case-health/silent
GET /api/case-health/stalled
GET /api/case-health/by-unit
GET /api/case-health/by-io
GET /api/case-health/:caseId
POST /api/cases/:caseId/activity
POST /api/case-health/review
```

Crime Intelligence

```
GET /api/crime/trends
GET /api/crime/by-district
GET /api/crime/by-category
GET /api/crime/change
GET /api/crime/anomalies
GET /api/crime/patterns
```

Geo

```
GET /api/geo/cases
GET /api/geo/hotspots
GET /api/geo/clusters
GET /api/geo/risk-map
GET /api/geo/districts
GET /api/geo/units
GET /api/geo/:lat/:lng/context
```

People / Network

```
GET /api/persons
GET /api/persons/:id
GET /api/persons/:id/cases
GET /api/persons/:id/network
GET /api/network/case/:caseId
GET /api/network/search
GET /api/network/clusters
```

Prediction

```
GET /api/predictions/crime
GET /api/predictions/hotspots
GET /api/predictions/district/:id
GET /api/predictions/explain/:id
GET /api/predictions/model-health
```

Intervention

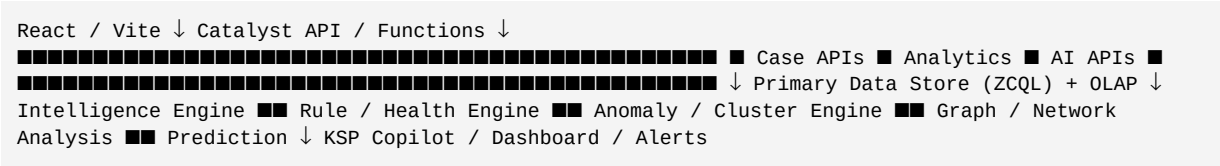
```
GET /api/interventions
POST /api/interventions
GET /api/interventions/:id
GET /api/interventions/:id/outcome
GET /api/interventions/effectiveness
```

AI

```
POST /api/ai/chat
POST /api/ai/query
POST /api/ai/summarize-case
POST /api/ai/explain-signal
POST /api/ai/explain-prediction
POST /api/ai/generate-briefing
POST /api/ai/generate-report
```

11. Catalyst Backend Architecture

Use the primary Catalyst Data Store for transactional/system-of-record operations and ZCQL CRUD. Use Catalyst's OLAP capability for read-heavy, multidimensional analytical queries. Server-side functions should own business logic, health calculations, signal generation and AI tool orchestration.



12. Real-Time vs Batch

Mode	Examples
Real-time	New FIR, critical incident, new attention signal, live patrol/event simulation.
Near-real-time	Dashboard KPIs, hotspot recalculation, operational metrics.
Batch	Predictive model refresh, monthly trend analysis, long-term intelligence reports.

13. Security & Governance

- Role-based and unit/district-scoped access: DGP/HQ, SP/DCP, SHO, IO, analyst.
- Audit AI queries and important analytical outputs.
- Separate personally identifying operational data from derived analytics where practical.
- Predictions and signals must be explainable and reviewable.
- Do not frame AI outputs as determinations of guilt, dangerousness or arrest decisions.

14. Implementation Roadmap

Phase	Build
1. Foundation	Populate master data and synthetic FIR/case universe; validate ZCQL and FK integrity.
2. Command Center	Karnataka map, KPIs, district drill-down, crime trends, filters, attention queue.
3. Case Intelligence	Case detail, timeline, case health, ageing, silent cases, investigation workload; add CaseActivity.
4. Intelligence	Hotspots, anomalies, repeat/entity graph, related cases, clusters.
5. Prediction	7/14/30-day risk, confidence, explainability and evaluation.
6. AI Copilot	Contextual Q&A, case summaries, briefings, explain/compare/drill-down actions.
7. Closed Loop	Interventions, outcome measurement and effectiveness.
8. Real Time	Event simulation, alerts, live map and resource integration.

15. Competitive Differentiator

Don't compete on 'we have a hotspot map' or 'we have an AI chatbot.' Those capabilities are already common in policing analytics. The stronger story is **Case-to-State Intelligence + Case Health + Action Feedback**.

Most systems answer: *Where is crime?*

KSP Intelligence Command should answer:

What happened → why it matters → which cases/workflows need attention → what should be reviewed → where should resources go → did the intervention work?

16. Evidence / Reference Links

The following current sources were checked while consolidating the plan. They are included as navigable references in the PDF.

Catalyst ZCQL documentation

<https://docs.catalyst.zoho.com/en/cloud-scale/help/zcql/introduction/>

Catalyst API — ZCQL query execution

<https://docs.catalyst.zoho.com/en/api/introduction/request-format/>

Catalyst Data Store

<https://docs.catalyst.zoho.com/en/cloud-scale/help/data-store/introduction/>

Catalyst OLAP Database

<https://docs.catalyst.zoho.com/en/cloud-scale/help/data-store/olap-database/introduction/>

Catalyst ZCQL Console

<https://docs.catalyst.zoho.com/en/cloud-scale/help/zcql/zcql-console/>

INTERPOL Project INSIGHT

<https://www.interpol.int/en/How-we-work/Criminal-intelligence-analysis/Projects/Project-INSIGHT>

INTERPOL Criminal Intelligence Analysis

<https://www.interpol.int/How-we-work/Criminal-intelligence-analysis>

INTERPOL Operational / Strategic Analysis Reports

<https://www.interpol.int/en/How-we-work/Criminal-intelligence-analysis/Our-analysis-reports>

Esri — Law Enforcement / Patrol Operations

<https://www.esri.com/en-us/industries/law-enforcement/strategies/patrol-operations>

Source/Scope Note

The police ERD and Karnataka crime-review materials discussed in the conversation establish the existing data model and aggregate reporting context. The proposed intelligence tables, health metrics, APIs, predictive layer, intervention model and UI are design recommendations for the hackathon, not claims about official Karnataka Police workflow or production policy. Any official priority, case-timeline or legal deadline should be validated against authoritative KSP/departmental rules before implementation.